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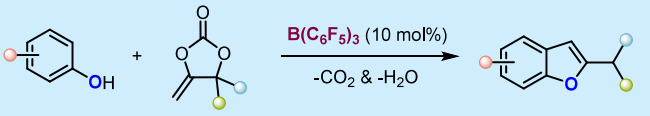
$B(C_6F_5)_3$-Catalyzed Decarboxylative Formation of Benzofurans via Intermolecular (3 + 2) Cyclization from Phenols

Xiao Shu, Yicheng He, and Wusheng Guo*

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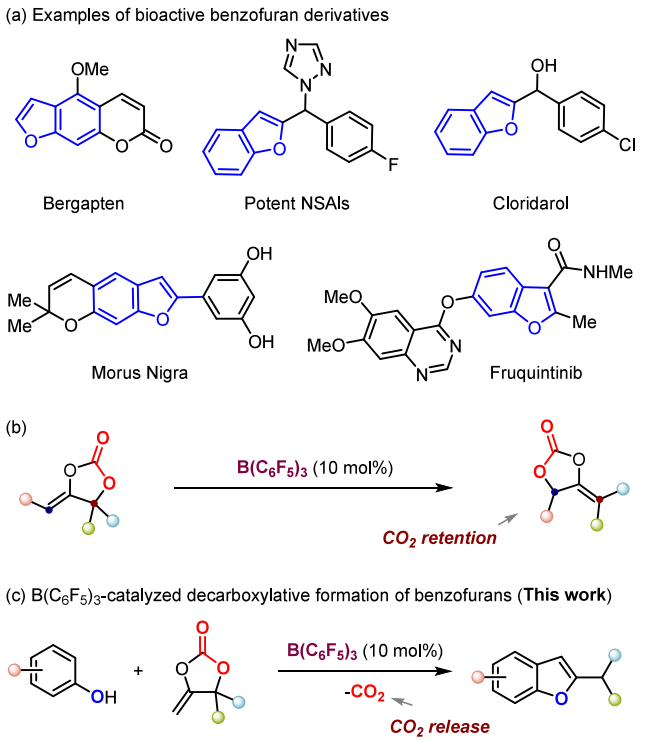
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ABSTRACT: We herein report unprecedented $B(C_6F_5)_3$ catalyzed intermolecular (3 + 2) cyclizations enabling versatile construction of benzofurans from readily available methylene cyclic carbonates and phenols. Such an organocatalytic process tolerated well different functional groups and only generated water and CO_2 as byproducts. The reactions proved to be scalable, and the corresponding products could be transformed into value-added benzofuran derivatives via simple experimental operations.



As a subclass of *O*-heterocycles, substituted benzofurans can be frequently found in many natural products, bioactive compounds, organic materials and pharmaceuticals (Scheme 1a).[1,2] Thus, the development of efficient processes for their versatile syntheses is highly desired and attracts considerable research interest.[3,4] Great success has been achieved via cyclization reactions between different reaction partners and prefunctionalized phenols.[3] In contrast, the synthesis of benzofurans from readily available phenols represents the most straightforward synthetic pathway. As such, respected efforts have been dedicated to the development of metal catalyzed formation of benzofurans from phenols,[4] albeit in most cases elevated reaction temperature (>100 °C)[4a−f] and a noble metal catalyst are required.[4b−g]

Scheme 1. Examples of Bioactive Benzofuran Derivatives and $B(C_6F_5)_3$-Catalyzed Transformation of Methylene Cyclic Carbonates



Methylene cyclic carbonates can be easily prepared from cyclization of propargylic alcohol and CO_2.[5] Their great synthetic application potentials have been demonstrated mainly based on transition metal catalyzed allylic chemistry[6] or Lewis acid-catalyzed intramolecular cyclization reactions.[7,5c] In continuation of our interests in the exploration of versatile reactivities of cyclic carbonates,[8,5b] we recently reported a $B(C_6F_5)_3$ catalyzed methylene cyclic carbonate editing approach (Scheme 1b);[9] such a CO_2-retaining process enables fast construction of the corresponding congeners of the starting cyclic carbonate. Unlike the reactivity noted previously,[9] we herein report an unprecedented Lewis acid $B(C_6F_5)_3$ catalyzed decarboxylative formation of benzofurans from phenols and methylene cyclic carbonates (Scheme 1c); such an intermolecular cyclization approach does not require the participation of noble metal catalyst and only generates CO_2 and water as byproducts.

We started our investigation by using the reaction between methylene cyclic carbonate **1a** and phenol **2a** as a model (Table 1). In the absence of any catalyst, no reaction occurred with the use of DCE as a solvent at room temperature (Table

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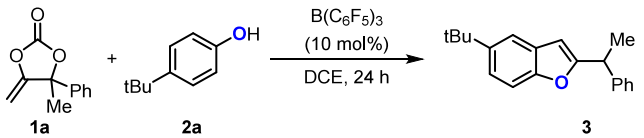


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Table 1. Selected Condition Screening Data toward the Formation of Benzofuran 3 from the Reaction between Cyclic Carbonate 1a and Phenol 2a[a]



1a 2a 3

Entry	Lewis Acid	Solvent	T/°C	3/%[b]
1	–	DCE	rt	NR
2	$BF_3{\cdot}Et_2O$	DCE	rt	50
3	$Cu(OTf)_2$	DCE	rt	10
4	$Sc(OTf)_3$	DCE	rt	<5
5	$FeCl_3$	DCE	rt	7
6	$Bi(OTf)_3$	DCE	rt	trace
7	$Zn(OTf)_2$	DCE	rt	16
8	CF_3COOH	DCE	rt	NR
9	$B(C_6F_5)_3$	DCE	rt	77
10	AgOTf	DCE	rt	22
11	$B(C_6F_5)_3$	DCM	rt	73
12	$B(C_6F_5)_3$	$CHCl_3$	rt	80
13	$B(C_6F_5)_3$	Acetone	rt	<5
14	$B(C_6F_5)_3$	THF	rt	<5
15	$B(C_6F_5)_3$	DCE	60	85
16	$B(C_6F_5)_3$	DCE	80	93 (91)[c]

[a]Reaction conditions: **1a** (0.12 mmol), **2a** (0.10 mmol), $B(C_6F_5)_3$ (10 mol %), DCE (0.3 mL), 80 °C, under protection of N_2, 24 h. [b]Yield determined by ^1H NMR yield using 2-Methylnaphthalene as an internal standard. [c]Isolated yield in parentheses.

1, entry 1). The addition of different Lewis acids as catalysts under otherwise identical conditions resulted in different reaction outcomes (entries 2−10). Among which, the use of $B(C_6F_5)_3$ as a catalyst gave rise to the best productivity (entry 9). Variation of the solvent did not improve the reaction remarkably (entries 11−14). To our delight, the reaction efficiency was greatly improved under higher reaction temperatures (entries 15−16).

With the optimized reaction conditions in hand, we then focused on investigating the substrate scope of this catalytic approach (Figures 1−2). To our delight, this organocatalytic system tolerated well different methylene cyclic carbonates featuring either electronically donating or withdrawing functional groups on the aryl substituent. A range of cyclic carbonates **1a−s** featuring different functional groups at the *para* (**4**−**11**), *meta* (**12**−**14**) or *ortho* positions (**15**−**16**) of the aryl substituents were reactive toward the target products in moderate to excellent yields. Cyclic carbonates featuring a naphthyl group reacted smoothly and gave rise to the corresponding benzofurans **20** and **21** in decent yields. The introduction of a benzyl (**18**) or cyclic fragment (**17**) in the benzofuran skeletons was also demonstrated. The equipment of the two phenyl substituents in the cyclic carbonate did not affect the reaction efficiency (**19**). In addition to NMR spectroscopy analysis, the molecular structure of benzofuran **7** was unambiguously further confirmed by X-ray diffraction analysis.[10] It was found that para- (**22**−**29**), meta- (**32**, **34**) or ortho-functionalized (**30**, **31**) phenols were all reactive toward the desired products. In contrast, the phenols featuring electronically donating groups are more reactive than those with electronically withdrawing functionalities. As expected, the use of meta-functionalized phenol as a reaction partner

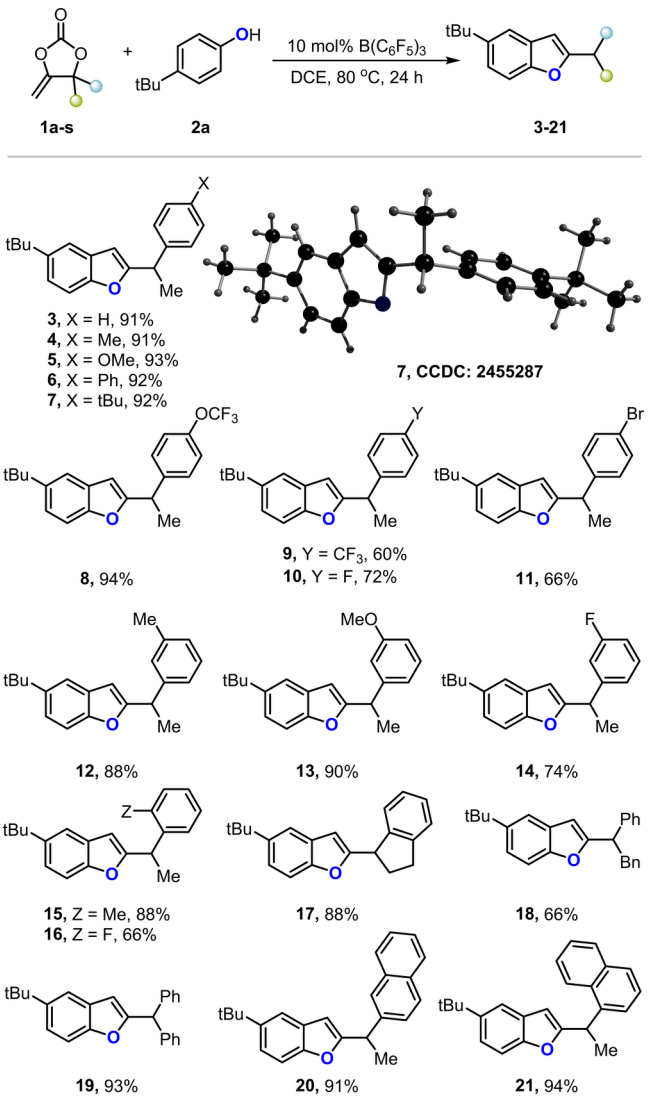


Figure 1. Reaction scope investigation in terms of methylene cyclic carbonates. Reaction conditions: cyclic carbonate **1** (0.12 mmol), phenol **2a** (0.10 mmol), $B(C_6F_5)_3$ (10 mol %), DCE (0.3 mL), 80 °C, under protection of N_2, 24 h. Isolated yields are reported.

gave rise to two regioisomers (**34** vs **34′**), while the use of phenols with bulkier meta-substituents (**32**, **33** vs **34**) resulted in improved regioselectivity, which is probably due to steric effects. The possibility for the introduction of a naphthyl group into the resulted benzofurans was also demonstrated, which may facilitate the use of these benzofurans in material science.[11] The successful input of bromo-functionalities in the benzofuran skeleton facilitated their usage in Suzuki coupling reactions, thus correspondingly creating much more potential in structural diversity. No reaction occurred with the use of aniline as the reaction partner under our optimized reaction conditions.

To further evaluate the practical application of this catalytic method, a gram-scale reaction was performed, and the reaction efficiency was not remarkably affected (Figure 3a). In addition to the reported application of benzofurans,[1,2] the synthetic value of these benzofuran products was further demonstrated (Figure 3b). For example, bromination of product **3** could be smoothly achieved under mild reaction conditions toward the formation of compound **35**.[12] Treatment of product **3** with

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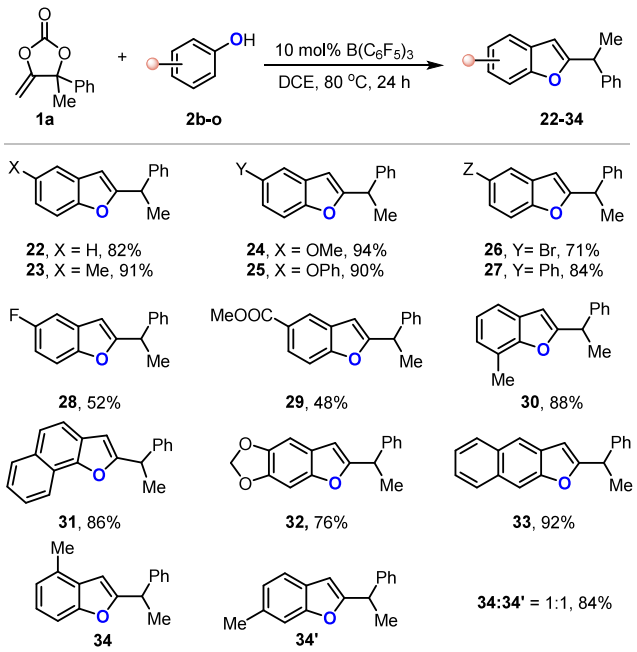


Figure 2. Reaction scope investigation in terms of phenols. Reaction conditions: cyclic carbonate **1a** (0.12 mmol), phenol **2** (0.10 mmol), $B(C_6F_5)_3$ (10 mol %), DCE (0.3 mL), 80 °C, under protection of N_2, 24 h. Isolated yields are reported.

Cl_2CHOMe and $TiCl_4$ led to formation of aldehyde **36**.[13] In order to gain more insight into the reaction mechanism (Figure 3c), we performed a control experiment with the use of commercially available compound **37** as starting material (Figure 3c). It was found that compound **37** was quantitatively converted into benzofuran **38** smoothly under the standard reaction conditions, indicating that a ketone like compound **37** probably worked as a key reaction intermediate. Based on our experimental observations and previous report,[9] a plausible reaction mechanism was proposed as depicted in Figure 3d. The interaction between cyclic carbonate **1a** and $B(C_6F_5)_3$ catalyst led to formation of intermediate **t1**, which is dynamically in equilibrium with intermediate **t2**. Decarboxylation of **t2** gave rise to the **t3** species. A subsequent intermolecular nucleophilic attack of phenol **2a** resulted in the formation of compound **t4**. Upon a cascade intramolecular nucleophilic attack and aromatization sequence, benzofuran **3** was produced and concomitantly generated water and CO_2 as byproducts. The hydrogen-bonding between intermediates **t3** and **2a** (boxed in Figure 3d) was believed to be beneficial for the carbon other than oxygen nucleophilic attack. As expected, the regioisomer derived from the nucleophilic attack of the more hindered carbon was not observed. This mechanism proposal is in accordance with the fact that the use of cyclic carbonate featuring an electronically donating group is more efficient than the ones featuring electronically withdrawing groups toward the formation of the target products.

In summary, we herein reported an unprecedented $B(C_6F_5)_3$ catalyzed intermolecular (3 + 2) cyclization approach enabling the versatile construction of benzofurans from readily available methylene cyclic carbonates and phenols. Such an organocatalytic process tolerated well different functional groups and only generated water and CO_2 as byproducts. The reactions proved to be scalable, and the corresponding products could

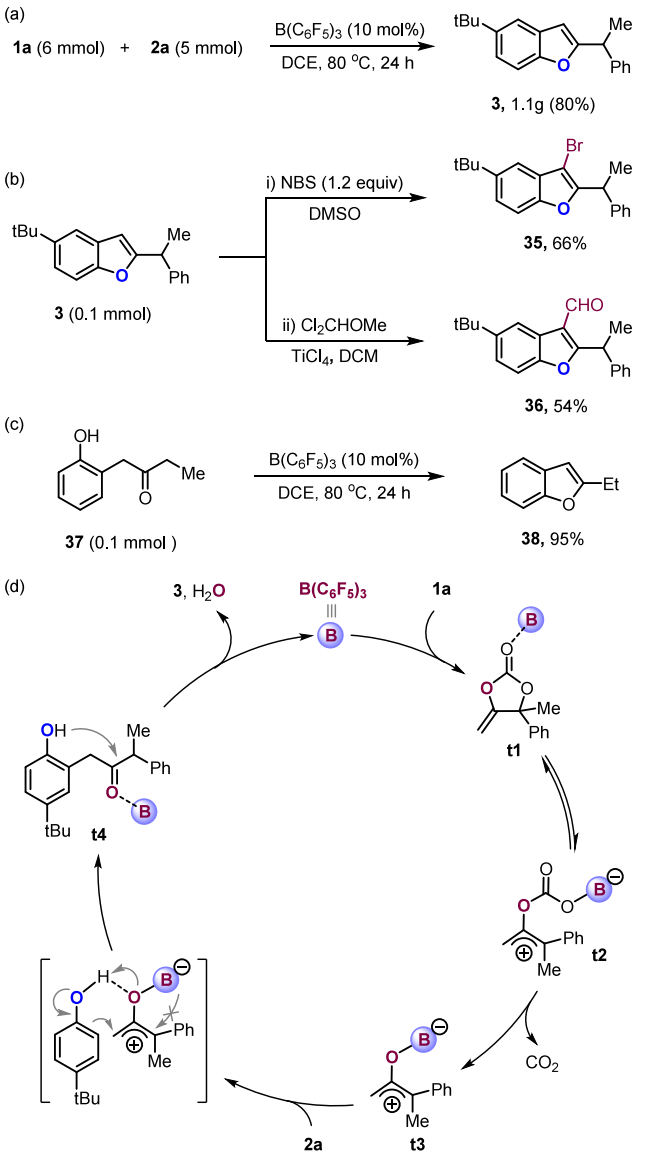


Figure 3. (a) Gram-scale reaction. (b) Synthetic transformations of product **3**. (c–d) Control experiments and mechanism proposal. Reaction conditions: (i) benzofuran **3** (0.1 mmol, 1 equiv), NBS (1.2 equiv), 35 °C, 12 h; (ii) benzofuran **3** (0.1 mmol, 1 equiv), Cl_2CHOMe (1.5 equiv), $TiCl_4$ (1.5 equiv), rt, 2 h. See the Supporting Information for experimental details.

be transformed into value-added benzofuran derivatives via simple operations.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.5c02208.

Experimental details and characterization data; crystallographic data for compounds **7** (PDF)

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Accession Codes

Deposition Number 2455287 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via the joint Cambridge Crystallographic Data Centre (CCDC) and Fachinformationszentrum Karlsruhe Access Structures service.

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Notes

The authors declare no competing financial interest.

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